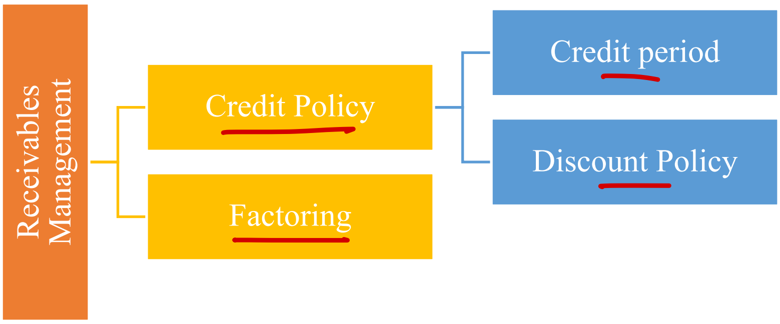
RECEIVABLES & PAYABLES MANAGEMENT

1. Management of Receivables

It refers to planning and controlling od debt owed to the firm from customer on account of credit sales. It is also known as trade credit management.



2. Credit period

It is the duration by which the amount becomes due and must be paid by the debtors.

Overdue - When debtors exceed credit period or don't' pay withing credit period

3. Statement of Evaluation of Credit Policy (Total Approach)

Particulars	Existing	Option I	Option II
Sales	--	--	--
Less: Variable cost	--	--	--
Less: Fixed cost	--	--	--
Less: Bad debts	--	--	--
Less: Discount	--	--	--
Profit before tax	--	--	--
Less: Tax	--	--	--
Profit after tax	--	--	--
Less: Opportunity cost on investment in debtors	--	--	--
Net Benefit	--	--	--

use if given in Ques.

4. Statement of Evaluation of Credit Policy (Incremental Approach)

Particulars	Existing	Option I	Option II
Sales	-- 100	-- 150	-- 180
Incremental Sales (A)		- 50	-- 80
Variable Cost	--	--	--
Incremental Variable Cost (B)		--	--
Fixed Cost	--	--	--
Incremental Fixed Cost (C)		--	--
Bad Debts	--	--	--
Incremental Bad Debts (D)		--	--
Discount	--	--	--
Incremental Discount (E)		--	--
Inc. Expected Profit before tax (A–B–C–D–E)		--	--
Less: Tax		--	--
Inc. Expected profit after tax (F)		--	--
Opportunity cost on investment in debtors	--	--	--
Inc. opportunity cost on investment in debtors (G)		--	--
Incremental Net Benefit (F – G)		--	--

5. Points to Remember (PTRs)

- Contribution = Sales – Variable Cost
- PV Ratio = $\frac{Contribution}{Sales} \times 100$ = 100 – Variable Cost Ratio
- If Average cost per unit is given than

  Fixed cost = (Average cost per unit – Variable cost per unit) × No. of units
- Opportunity cost on investment in debtors = $\boxed{(VC + FC) \times \frac{ACP}{365/52/12}}$ × Rate
- Opportunity cost on investment in debtors = $\boxed{(VC + FC) \times \frac{1}{DTR}}$ × Rate
- Special care to be taken regarding opportunity cost rate whether it is before tax or after tax and accordingly tax treatment should be done.

  opp. cost BT (1 - t) = AT
- ACP = (Days)(W1) + (Days)(W2) +……..+ (Days)(Wn)

6. Discount Policy

Meaning of discount terms 1/10 Net 40

Get 1% discount if paid within 10 days else pay withing 40 days without discount.

$$\text{Effective/Implied Annual rate of Discount} = \frac{Discount}{(100-Discount)} \times \frac{365}{No.\ of\ days\ of\ prepayment} \times 100$$

Question – 1	

SK corporation is considering relaxing its present credit policy and is in the process of evaluating two proposed policies. Currently, the firm has annual credit sales of ₹ 50 lakhs and accounts receivable turnover ratio of 4 times a year. The current level of loss due to bad debts is ₹ 1,50,000. The firm is required to give a return of 25% on the investment in new accounts receivables. The company's variable costs are 70% of the selling price. Given the following information, which is the better option?

	Present Policy	Policy I	Policy II
Annual credit sales	₹ 50,00,000	₹ 60,00,000	₹ 67,50,000
Accounts Receivables turnover ratio	4 times	3 times	2.4 times
Bad debts losses	1,50,000	3,00,000	4,50,000

Solution

Statement of Credit Policy Evaluation

Particulars	Existing	Policy I	Policy II
Credit Sales	50,00,000	60,00,000	67,50,000
Less: Variable cost @ 70%	35,00,000	42,00,000	47,25,000
Less: Bad Debts	1,50,000	3,00,000	4,50,000
Expected Profit	13,50,000	15,00,000	15,75,000
Less: Opportunity cost	35,00,000 × (1/4) × 25% = 2,18,750	42,00,000 × (1/3) × 25% = 3,50,000	47,25,000 × (1/2.4) × 25% = 4,92,188
Net Benefit	11,31,250	$\boxed{11,50,000}$	10,82,812

Net benefit is higher in case of Policy I, thus Policy I is better.

Question – 2	

AB Enterprises deals in hardware materials having current turnover ₹30 Lakhs per annum. All sales are on credit and average collection period is 30 days with zero bad debts. The customers are requesting to increase the credit period. As a result of increase in credit period sales will also increase. Other information is as under:

VC Ratio = $\frac{50}{100} \times 100 = 50\%$

Credit policy	Increase in collection period (days)	Increase in sales (₹)	Bad debts anticipated
A	30 + 15 = 45	30L + 3,00,000 = 33L	× 1% = 33000
B	30 + 30 = 60	30L + 5,00,000 = 35L	× 3.5% = 122500

The Selling price is ₹100/- per unit. Variable cost per unit is ₹50/- and fixed cost is ₹5,00,000. Required rate of return on additional investment is 20%. Creditors for variable cost are ready to give 15 days extra credit for the additional cost incurred. Assume a 360 days year.

You are required to analyze the present and proposed credit policies using the "Total Approach" method and recommend the credit policy to be adopted.

$1.5L \times \frac{15}{360} = 6250$ $\quad$ $2.5L \times \frac{15}{360} = 10417$

Solution

A. Statement showing the Evaluation of Credit Policies (Total Approach)

	Particulars	Present Credit Policy	Proposed Credit Policy	
			A	B
	Credit Period (in days)	30	45	60
	Units sold	30,000	33,000	35,000
		₹	₹	₹
A	**Expected Profit:**			
	(a) Credit Sales @ ₹100 per unit	30,00,000	33,00,000	35,00,000
	(b) Total Cost other than Bad Debts			
	(i) Variable Costs @ ₹50 per unit	15,00,000	16,50,000	17,50,000
	(ii) Fixed Costs	5,00,000	5,00,000	5,00,000
		20,00,000	21,50,000	22,50,000
	(c) Bad Debts	–	33,000	1,22,500
	(d) Expected Profit [(a) – (b) – (c)]	**10,00,000**	**11,17,000**	**11,27,500**
B	**Opportunity Cost of Investments in Receivables (w.n. i)**	**33,333**	**52,500**	**72,917**
C	**Net Benefits (A – B)**	**9,66,667**	**10,64,500**	**10,54,583**

Recommendation: The Proposed Policy A (i.e. increase in collection period by 15 days or total 45 days) should be adopted since the net benefits under this policy are higher as compared to other policies.

Working Notes:

(i) Calculation of Opportunity Cost of Average Investment in Receivables

	Particulars	Present Credit Policy	Proposed Credit Policy	
			A	B
	Credit Period (in days)	30	45	60
		₹	₹	₹
(a)	Cost of Sales (Variable Cost + Fixed Cost)	20,00,000	21,50,000	22,50,000
(b)	Average Debtors = Cost of Sales × (credit period)/360	$\frac{20L}{360} \times 30 =$ 1,66,667	$\frac{21.5L}{360} \times 45 =$ 2,68,750	3,75,000 $= \frac{22.50L}{360} \times 60$
(c)	Average Creditors for extra Variable Cost) × 15/360]	–	6,250	10,417
(d)	**Average Investment in Receivables or Net Working Capital = (b) – (c)**	$\boxed{1,66,667}$ ↓ 20%	$\boxed{2,62,500}$ ↓ 20%	$\boxed{3,64,583}$ ↓ 20%
(e)	**Opportunity Cost @20% of average Investments in Receivables**	33,333	52,500	72,917

Question – 3	

As a part of the strategy to increase sales and profits, the sales manager of a company proposes to sell goods to a group of new customers with 10% risk (B/Dts) of non-payment. This group would require one and a half months credit and is likely to increase sales by ₹ 1,00,000 p.a. Production and Selling expenses amount to 80% of sales and the income-tax rate is 50%. The company's minimum required rate of return (after tax) is 25%. Should the sales manager's proposal be accepted? Analyse.

Also compute the degree of risk of non-payment that the company should be willing to assume if the required rate of return (after tax) were (i) 30%, (ii) 40% and (iii) 60%.

Req. Return = (Sales − COS − B/Dts)(1−t)

$80000 \times \frac{1.5}{12} \times 30\% = (1L - 0.8L - B/Dts)(1-0.5)$

$3000 = (20000 - B/Dts)(0.5)$

$6000 = 20000 - B/Dts$

$B/Dts = 14000$

$B/Dts\ \% = \frac{14000}{1L} = 14\%$

Solution

(a) Statement of Credit Policy Evaluation

Particulars	Amount
Increase in Sales	1,00,000
Less: Cost of sales @ 80%	(80,000)
Less: Bad Debts (1,00,000 × 10%)	(10,000)
Expected Profit Before Tax	10,000
Less: Tax @ 50%	(5,000)
Expected Profit After Tax	5,000
Less: Opportunity cost (80,000 × 1.5/12 × 25%)	(2,500)
Net Benefit after tax	$\boxed{2,500}$

Due to higher net benefit, it is recommended to accept the proposal.

(i) Required return after tax = (Sales – Cost of sales – Bad Debts)(1 – t)
80,000 × 1.5/12 × 30% = (1,00,000 – 80,000 – Bad Debts)(1 – 0.50)
Bad Debts = ₹ 14,000
Degree of risk of non-payment = 14,000 ÷ 1,00,000 = 14%

(ii) Required return after tax = (Sales – Cost of sales – Bad Debts)(1 – t)
80,000 × 1.5/12 × 40% = (1,00,000 – 80,000 – Bad Debts)(1 – 0.50)
Bad Debts = ₹ 12,000
Degree of risk of non-payment = 12,000 ÷ 1,00,000 = 12%

(iii) Required return after tax = (Sales – Cost of sales – Bad Debts)(1 – t)
80,000 × 1.5/12 × 60% = (1,00,000 – 80,000 – Bad Debts)(1 – 0.50)
Bad Debts = ₹ 8,000
Degree of risk of non-payment = 8,000 ÷ 1,00,000 = 8%

Disc. = $12L \times \frac{50}{100} \times \frac{1}{100} =$
New D. = $16L \times \frac{80}{100} \times \frac{2}{100} =$

Question – 4

A company is presently having credit sales of ₹ 12 lakh. The existing credit terms are 1/10, net 45 days and average collection period is 30 days. The current bad debts loss is 1.5%. in order to accelerate the collection process further as also to increase sales, the company is contemplating liberalization of its existing credit terms to 2/10, net 45 days. It is expected that sales are likely to increase by 1/3 of existing sales, bad debts increase to 2% of sales and average collection period to decline to 20 days. The contribution to sales ratio of the company (VC = 78%) is 22% and opportunity cost of investment in receivables is 15 per cent (pre-tax). 50 percent and 80 percent of customers in terms of sales revenue are expected to avail cash discount under existing and liberalization scheme respectively. The tax rate is 30%. Advise, should the company change its credit terms? (assume 360 days in a year).

Solution **Statement of Evaluation of Discount Policy (Total Approach)**

Particulars	Existing	Proposed
Sales	12,00,000	12,00,000 + (1/3) = 16,00,000
Less: Variable cost @ 78%	12,00,000 × 78% = 9,36,000	16,00,000 × 78% = 12,48,000
Less: Discount	12,00,000 × 50% × 1% = 6,000	16,00,000 × 80% × 2% = 25,600
Less: Bad Debts	12,00,000 × 1.5% = 18,000	16,00,000 × 2% = 32,000
Profit before tax	2,40,000	2,94,400
Less: Tax @ 30%	72,000	88,320
Profit after tax	1,68,000	2,06,080
Less: Opportunity cost* 15(1- 0.30) = 10.50%	$9,36,000 \times \frac{30}{360} \times 10.5\% =$ 8,190	$12,48,000 \times \frac{20}{360} \times 10.5\% =$ 7,280
Net Benefit	1,59,810	Inc. 1,98,800

Recommend to accept the proposal as it has more benefit then existing situation.

*Opportunity cost rate after tax = 15 × (1 – 0.30) = 10.5%

Question – 5	

Mr. S is regular customers of SK Ltd. and have approached the sellers for extension of a credit facility for enabling them to purchase goods from SK Ltd. On an analysis of past performance and on the basis of information supplied, the following pattern of payment schedule emerges in regard to Mr. S:

Schedule		Pattern
At the end of 30 days	–	15% of the bill
At the end of 60 days	–	34% of the bill
At the end of 90 days	–	30% of the bill
At the end of 100 days	–	20% of the bill
Non-recovery (B/D+S)	–	1% of the bill

ACP = (30)(0.15) + (60)(0.34) + (90)(0.30) + (100)(0.20) = 71.9 days

15L × 145/150 = (14.50L)

15L

(5000)

15L × 1% = (15000)

30000

14.50L × 71.9/360 × 24% =

Mr. S want to enter into a firm commitment for purchase of goods of ₹ 15 lakhs in 2021, deliveries to be made in equal quantities on the first day of each quarter in the calendar year. The price per unit of commodity is ₹ 150 on which a profit of ₹ 5 per unit is expected to be made. It is anticipated by SK Ltd., that taking up of this contract would mean an extra recurring expenditure of ₹ 5,000 per annum. If the opportunity cost of funds in the hands of SK Ltd. is 24% per annum would you as the finance manager of the seller recommend the grant of credit to Mr. S? Workings should form part of your answer. Assume year of 360 days.

Solution

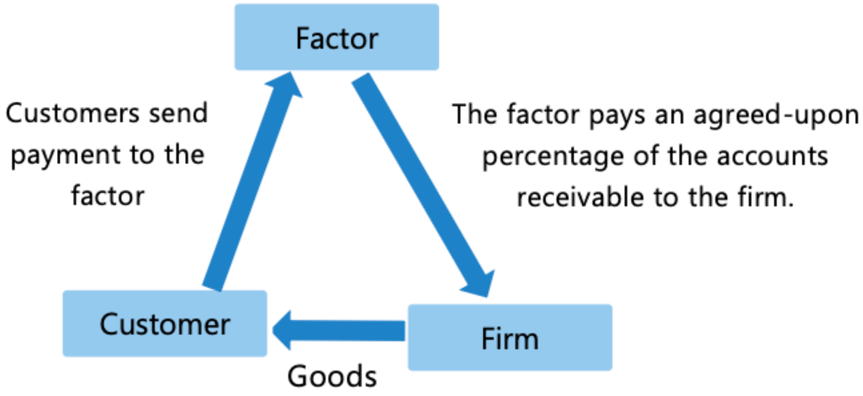
ACP = (30days ×15%) + (60days ×34%) + (90 days × 30%) + (100 days × 20%) = 71.9 days

Statement showing evaluation of credit policy

Particulars	Amount
Sales (10,000 × 150)	15,00,000
Less: Variable cost (10,000 × 145)	(14,50,000)
Less: Extra cost expenditure	(5,000)
Less: Bad Debts (15,00,000 × 1%)	(15,000)
Expected Profit	30,000
Less: Opportunity cost (14,55,000 × 71.9/365 × 24%)	(68,788)
Net Benefit	(38,788)

Not recommended to accept the policy as it has negative benefit.

7. Factoring



It refers to outright sale of accounts receivables to a factor or a financial agency. A factor is a firm that acquires the receivables of other firms. The factoring lays down the conditions of the sale in a factoring agreement. The factoring agency bears the risk of collection and services the accounts for a fee.

Recourse: In case factor is unable to collect the amount from receivables then, factor can turn back the same to the organizations for resolution.

Non-Recourse: The factor bears the ultimate risk of loss in case of default and hence in such cases they charge higher commission.

Statement of Evaluation of Factoring Proposal

Particulars	Amount
Saving in Bad debts	--
Saving in administration cost	--
Interest / Opportunity cost saving [Cost of credit sales × Int. rate × $\frac{(Present\ ACP - New\ ACP)}{360}$]	--
Any other savings	--
Total Savings (A)	--
Factoring Commission	--
Interest cost on advance	--
Any other cost	--
Total Costs (B)	--
Net Benefit or Cost (A – B)	--

8. Points to Remember (PTRs)

(A) Factoring commission = Annual credit sales × Commission %

(B) Calculation of amount of advance

Particulars	Amount
Annual credit sales	--
(-) Factor Reserve	--
(-) Factoring Commission	--
Amount available for advance	--
(-) Interest [Amount available for advance × $\frac{\textit{Collection period}}{360}$ × Int. rate]	--
Amount of Advance	--

*Commission and Interest are payable in advance.

(C) Rate of effective cost of factoring = $\frac{\textit{Net annual cost of factoring}}{\textit{Amount of advance/ Amount available for advance}} \times 100$

Net annual cost of factoring = Cost of Factoring – Savings due to Factoring

(D) Compare rate of effective cost of factoring with bank interest rate to take decision.

Question – 6	

A factoring firm has offered a company to buy its accounts receivables. The relevant information is given below:

(i) The current average collection period for the company's debt is 80 days and ½% of debtors default. The factor has agreed to pay over money due to the company after 60 days and it will suffer all the losses of bad debts also.

B/DB

20 days reduce

(ii) Factor will charge commission @2%.

(iii) The company spends ₹ 1,00,000 p.a. on administration of debtor. These are avoidable costs.

(iv) Annual credit sales are ₹ 90 lakhs. Total variable costs is 80% of sales. The variable costs is 80% of sales. The company's cost of borrowing is 15% per annum. Assume 365 days in a year.

Should the company enter into agreement with factoring firm?

Benefit

B/DB = 90L × 0.5% = 45000

Adm. = 100 000

Opp. Cost = 90L × 80% × $\frac{(80-60)}{365}$ × 15% = 59178

204178

Costs

Comm. [2% × 90L] 1.80L

1.80L

Net Benefit = 24178

Solution

Presently, the debtors of the company pay after 80 days. However, the factor has agreed to pay after 60 days only. So, the investment in debtors will be reduced by 20 days. The annual charge in cash flows through entering into a factoring agreement is:

Particulars	₹
Factoring commission (90,00,000 × 2%)	(1,80,000)
Administration cost saved	1,00,000
Bad debts saved (90,00,000 × 0.50%)	45,000
Interest saving [{(90,00,000 × 80/360) – (90,00,000 × 60/360)} × 80% × 15%]	59,178
Net Benefit	**24,178**

Recommended to enter into factoring agreement as it will provide annual benefit of ₹ 24,178.

Question – 7

A firm has credit sales of ₹360 lakhs and its average collection period is 30 days. The financial controller estimates, bad debt losses are around 2% of credit sales. The firm spends ₹1,40,000 annually on debtors administration. This cost comprises of telephonic and fax bills along with salaries of staff members. These are the avoidable costs. A factoring firm has offered to buy the firm's receivables. The factor will charge 1% commission and will pay an advance against receivables on an interest @15% p.a. after withholding 10% as reserve. Analyse what should the firm do? Assume 360 days in a year.

Solution

Evaluation of Factoring Proposal

Particulars		(₹)
(A)	**Savings to the firm**	
	Bad Debts saved (0.02 × 360 lakhs)	₹7,20,000
	Cost of credit administration	₹1,40,000
	Total (A)	₹8,60,000
(B)	**Costs of factoring:**	
	Factoring Commission (30,000 × 360/30)	₹3,60,000
	Interest cost (33,375 × 360/30)	₹4,00,500
	Total (B)	₹7,60,500
(C)	**Net Benefit to the Firm: (A-B)**	₹99,500

Since the savings to the firm exceeds the cost to the form exceeds the cost to the firm on account of factoring, therefore the proposal is acceptable.

Annual Credit Sales	₹360 lakhs
Average level of receivables = ₹360 lakhs × 30/360	₹30,00,000
(-) Commission = 1% of ₹30 lakhs	₹30,000
(-) Reserve = 10% of ₹30 lakhs	₹3,00,000
Amount Available for Advance	₹26,70,000
Less: Interest @ 15% p.a. for 30 days	₹33,375
Net Amount of Advance available.	₹26,36,625

Question – 8	

Following is the sales information in respect of Bright Ltd:

Annual Sales (90 % on credit)	₹7,50,00,000
Credit period	45 days
Average Collection period	70 days
Bad debts	0.75%
Credit administration cost (out of which 2/5th is avoidable)	₹18,60,000 × 2/5 = 7.44 L

A factor firm has offered to manage the company's debtors on a nonrecourse basis at a service charge of 2%. Factor agrees to grant advance against debtors at in interest rate of 14% after withholding 20% as reserve. Payment period guaranteed by factor is 45 days. The cost of capital of the company is 12.5%. One time redundancy payment of ₹50,000 is required to be made to factor.

Calculate the effective cost of factoring to the company. (Assume 360 days in a year)

Solution

Evaluation of Factoring Proposal

Particulars	**(₹)**	**(₹)**
(A) Savings due to factoring		
Bad Debts saved	0.75% × 7.5 crores × 90%	₹5,06,250
Administration cost saved	18.6 lakhs × 2/5	₹7,44,000
Interest saved due to reduction in average collection period	7.5 crores × 90% × (70-45)/ 360 × 12.5%	₹5,85,937.5
Total	A	₹18,36,187.5
(B) Costs of factoring:		
Service charge [168750 × 360/45]	7.5 crores × 90% × 2%	₹13,50,000
Interest cost	₹1,15,171.875 × 360/45	₹9,21,375
Redundancy Payment		₹50,000
Total	B	₹23.21,375
(C) Net Annual cost to the Firm: (A-B)		₹4,85,187.5
Rate of effective cost of factoring	₹4,85,187.5/ ₹64,66,078.125 × 100	7.504%

Advice: Since the rate of effective cost of factoring is less than the existing cost of capital, therefore, the proposal is acceptable.

Credit Sales = ₹7.5 crores × 90%	₹6,75,00,000
Average level of receivables = ₹6.75 crores × 45/360	₹84,37,500
(-) Service charge = 2% of ₹84,37,500	₹1,68,750
(-) Reserve = 20% of ₹84,37,500	₹16,87,500
Amount Available for Advance	₹65,81,250
Less: Interest @ 14% p.a. for 45 days	₹1,15,171,875
Net Amount of Advance available.	₹64,66,078.125

6581250 × 14/100 × 45/360

Question – 9	

Sukrut Limited has annual credit sales of ₹75,00,000/-. Actual credit terms are 30 days, but its management of receivables has been poor, and the average collection period is about 60 days. Bad debt is 1 per cent of total sales.

A factor has offered to take over the task of debt administration and credit checking, at an annual fee of 1.5 per cent of credit sales. Sukrut Limited estimates that it would save ₹45,000 per year in administration costs as a result. Due to the efficiency of the factor, the average collection period would come back to the original credit offered of 30 days and bad debts would come to 0.5% on recourse basis.

The factor would pay net advance of 80 percent to the company at an annual interest rate of 12 per cent after withholding a reserve of 10%. Sukrut Limited is currently financing its receivables from an overdraft costing 10 per cent per year and will continue to finance the balance fund needed (which is not financed by factor) through the overdraft facility

If occurrence of credit sales is throughout the year, COMPUTE whether the factor's services should be accepted or rejected. Assume 360 days in a year.

Solution

Evaluation of Factoring Proposal -

	PARTICULARS	₹	₹
(A)	**Savings (Benefit) to the firm**		
	Administration Cost	45,000	45,000
	Bad Debts Cost (On Recourse basis) In House – 75 lakhs X 1% Factoring – 75 lakhs X 0.5% **Net Savings in bad debts cost**	(75 lakhs X 0.5%)	37,500
	Cost of Carrying Debtors Cost	**(WN – 1)**	1,06,750
	TOTAL		**1,89,250**
(B)	**Cost to the Firm:**		
	Factor Commission [Annual credit Sales × % of Commission]	75 lakhs X 1.5%	1,12,500
	Interest Cost on Net advances	**(See WN – 1)**	53,100
	TOTAL		**1,65,600**
(C)	**Net Benefits to the Firm (A – B)**		**23,650**

Advice: Since the savings to the firm exceed the cost due to factoring, the proposal is acceptable.

WN-1 : Calculation of Savings in Interest Cost of Carrying Debtors

(I) In house Management:

Interest Cost = Credit Sales X Avg Collection Period / 360 X Interest (%) p.a

= 75,00,000 x 60/360 x 10%

= 1,25,000

(II) If Factoring services availed: If factoring services are availed, then Sukrut Limited must raise the funds blocked in receivables to the extent which is not funded by the factor (i.e amount of factor reserve **(+)** amount of factor commission for 30 days **(+)** 20% of net advances)

Calculation of Net Advances to the firm -

Debtors = 75 lakhs x 30/360 = 6,25,000

(-) Factor Reserve = 10% of above = (62,500)

(-) Factor Commission = 1.5% of Debtors = (9,375)

Net Advance = 5,53,125

Advance from Factor = 5,53,125 x 80% = 4,42,500

Int cost on Advance from Factor = 4,42,500 x 12% = 53,100

Now, the amount that is not funded by the factor (6,25,000 - 4,42,500) needs to be funded by Sukrut Limited from overdraft facility at 10%

Therefore, Int cost on Overdraft (Cost of carrying debtors) **= 1,82,500 x 10% = 18,250**

Net Savings in Interest Cost of Carrying Debtors = 1,25,000 (-) 18,250 = 1,06,750

Question – 10	

The Alliance Ltd. a Petrochemical sector company had just invested huge amount in its new expansion project. Due to huge capital investment, the company is in need to an additional ₹ 1,50,000 in working capital immediately. The finance Manager has determined the following three feasible sources of working capital funds:

$\frac{(2L \times 15\%)}{(2L \times 90\%)} \times 100 = 16.67\%$

(i) Bank Loan: The company's bank will lend ₹ 2,00,000 at 15%. A 10% compensating balance will be required, which otherwise would not be maintained by the company.

(ii) Trade Credit: The company has been offered credit terms from its major supplier of 3/30, net 90 for purchasing raw materials worth ₹ 1,00,000 per month. $\frac{3}{97} \times \frac{360}{80} \times 100 = 18.56\%$

(iii) Factoring: A factoring firm will buy the company's receivables of ₹ 2,00,000 per month, which have a collection period of 60 days. The factor will advance up to 75% of the face value of the receivables at 12% on an annual basis. The factor will also charge commission of 2% on all receivables purchased. It has been estimated that the factor's services will save the company a credit department expenses and bad debt expense of ₹ 1,250 and ₹ 1,750 per month respectively.

On the basis of annual percentage cost, advise which alternative should the company select? Assume 360 days year.

$(2\% \times 2L \times 12) + (2L \times 75\% \times 12\%) - (1250 \times 12) - (1750 \times 12) = 30000$

Fact Cost % $= \frac{30000}{150000} \times 100 = 20\%$

Solution

(i) Bank Loan

Loan amount = ₹ 2,00,000; Usable amount = ₹ 2,00,000 × 90% = ₹ 1,80,000

Real annual cost = $\frac{Interest}{Funds\ available} \times 100 = \frac{(2,00,000 \times 15\%)}{1,80,000} \times 100$ = 16.67% p.a.

(ii) Trade Credit

In this case, discount will not be taken which will cost = $\frac{3}{(100-3)} \times \frac{360}{30} \times 100$ = 18.56% p.a.

(iii) Factoring

Amount available for advance = 2,00,000 × 75% = ₹ 1,50,000

Commission charges = 2,00,000 × 12 × 2% = ₹ 48,000

Annual interest cost = ₹ 1,50,000 × 12% = ₹ 18,000

Savings per year = (1,250 + 1,750) × 12 = ₹ 36,000

Net factoring cost per year = 48,000 + 18,000 – 36,000 = ₹ 30,000

Effective cost of factoring = $\frac{30,000}{1,50,000} \times 100$ = 20% p.a.

Advise: The company should select bank loan alternative as it has the lowest annual cost.

9. Pledging

It refers to the use of firm's receivables to secure a short-term loan. The lender scrutinizes the quality of the account receivables, selects acceptable accounts, creates a lien on the collateral and fixed the percentage of financing receivables which ranges around 50% to 90%.

Question – 11	

Nirmoh Limited wants to avail short-term loan from the bank. However, bank grants short term loan by keeping the collateral in the form of accounts receivable. A bank is analyzing the receivables of Nirmoh Limited to identify acceptable collateral for a short-term loan.

The current policy of the company is 3/10 net 40. Bank will lend only to the extent of 90% of acceptable receivables at an interest rate of 12% only if both the conditions mentioned below are fulfilled. Bank will keep a reserve of 5% for cash discount & returns

(a) Customers are not currently overdue for more than 5 days to the net period + 40 = $\boxed{45}$

(b) Average aging (payment period) of the customer should not exceed 15 days past the net period. + 40 = $\boxed{55}$

If any of the above conditions are not fulfilled, the bank will lend 65% of the receivables subject to a reserve of 15% and the interest rate will be charged at 15% on such accounts. The corporate tax rate applicable is 25%.

On the scrutiny of all the receivables, following are the acceptable receivables considered for lending-

Accounts	Amount (₹)	Outstanding in Days since invoiced (45)	Average Aging (payment period) in Days (55)
DR 01 90%	50,000	37	40
DR 02 90%	25,000	25	48
DR 03	65% 1,20,000	47	49
DR 04 65%	72,000	10	56
DR 05 90%	45,000	30	30
DR 06 90%	1,75,000	39	50
DR 07 65%	19,000	55	25
DR 08	90% 54,000	44	54
DR 09	90% 1,05,000	15	25
DR 10	65% 37,000	22	75

You are required to calculate:

(a) Total amount lend by the bank

(b) Effective interest cost (%) to the company

Solution

(a) Condition (a) says that accounts shouldn't be overdue for more than 5 days to the net period. In other words, it means those accounts who are overdue by 45 days (40 days + 5 additional days), will not fulfil condition a) and thus will not be eligible for 90% lending.

Therefore, from the above, we can see that Accounts DR 03 & DR 07 are overdue for more than 45 days and hence will not be eligible for 90% lending.

Condition (b) says that average receivables ageing (payment period) should not exceed 15 days to the net period i.e. it should not exceed 55 days (40 days + 15 days = 55 days). Therefore, from the above, we can see that Accounts DR 04 & DR 10 has an ageing of more than 55 days. Hence, they would also not be eligible for 90% lending.

Amount of Bank Lending:

Accounts	Bank Lending at 90%	Bank Lending at 65%
DR 01	50,000	-
DR 02	25,000	-
DR 03	-	1,20,000
DR 04	-	72,000
DR 05	45,000	-
DR 06	1,75,000	-
DR 07	-	19,000
DR 08	54,000	-
DR 09	1,05,000	-
DR 10	-	37,000
Total	**4,54,000**	**2,48,000**
(-) Reserve	22,700 (4,54,000 × 5%)	37,200 (2,48,000 × 15%)

Net	**4,31,300**	**2,10,800**
Loan	**3,88170** 90%	**1,37,020** 65%

Total short-term loan granted by the bank = ₹5,25,190

(b) Interest at 12% (on 90% lending) = 3,88,170 × 0.12 = ₹46,580.4

Interest at 15% (on 65% lending) = 1,37,020 × 0.15 = ₹20,553

Total Interest = ₹67,133.4

Effective Interest Cost (%) = $\frac{Interest(1-t)}{Total\ short-term\ loan} = \frac{67{,}133.4(1-0.25)}{5{,}25{,}190} = 9.59\%$

10. Management of Payables

Trade creditor is a spontaneous / short term source of finance in the sense that it arises from ordinary business transaction. But it is also important to look after your creditors – slow payment by you may create ill-feeling and your supplies could be disrupted and also create a bad image for your company.

Cost of not taking discount on annual basis = $\frac{Discount}{(100-Discount)} \times \frac{365}{No.\ of\ days\ of\ prepayment} \times 100$

Question – 12	

A Ltd. is in the manufacturing business and it acquires raw material from X Ltd. on a regular basis. As per the terms of agreement the payment must be made within 40 days of purchase. However, A Ltd. has a choice of paying ₹ 98.50 per ₹ 100 it owes to X Ltd. on or before 10th day of purchase.

Required to examine whether A Ltd. should accept the offer of discount assuming average billing of A Ltd. with X Ltd. is ₹ 10,00,000 and an alternative investment yield a return of 15% and company pays the invoice.

$\frac{1.5}{98.5} \times \frac{360}{30} \times 100 = 18.53\%$

Solution

Annual benefit of accepting the discount = $\frac{1.5}{100-1.5} \times \frac{365}{40-10} \times 100 = 18.53\%$

Annual cost = Opportunity cost of foregoing interest on investment = 15%

If average invoice amount is ₹ 10,00,000.

	If discount is	
	Accepted (₹)	**Not accepted (₹)**
Payment to supplier	9,85,000	10,00,000
Return on investment of 9,85,000 for 30 days [9,85,000 × (30/365) × 15%]	-	(12,144)
	9,85,000	9,87,856

Thus, from above table it can be seen that it is cheaper to accept the discount.

Question – 13

The Dolce Company purchases raw material on terms of 2/10 net 30. A review of the company's records by the owner, Mr. Gautam, revealed that payments are usually made 15 days after purchases are made. When asked why the firm did not take advantage of its discounts, the accountant, Mr. Rohit, replied that it cost only 2 per cent of these funds, whereas a bank loan would cost the company 12 per cent.

$\frac{2}{98} \times \frac{360}{5} \times 100 = 146.94\%$

(a) Analyse what mistake is Rohit making?

(b) If the firm could not borrow from the bank and was force to resort to the use of trade credit funds, what suggestions might be made to Rohit that would reduce the annual interest cost? Identify.

$\frac{2}{98} \times \frac{360}{20} \times 100 =$

Solution

(a) Rohit's argument of comparing 2% discount with 12% bank loan rate is not rational as 2% discount can be earned by making payment 5 days in advance i.e. withing 10 days rather 15 days as payments are made presently. Whereas 12% bank loan rate is for a year.

Assume that the purchase value is ₹100, the discount can be earned by making payment within 10 days is ₹2, therefore net payment would be ₹98 only.

Annualized benefit $= \frac{2}{98} \times \frac{365\ days}{5\ days} \times 100 = 149\%$

This means cost of not taking cash discount is 149%.

(b) If the bank loan facility could not be available, then in this case the company could resort to utilise maximum credit period as possible.

Therefore, payment should be made in 30 days to reduce the interest cost.